

NO-CHURN TELECOM

Customer Churn Analysis Report

Project Ref: PM-PR-0017 | DataMites™

Executive Summary

This report presents the findings of a Machine Learning project undertaken for No-Churn Telecom to identify, predict, and flag customers at high risk of churning to competitors.

Best Model	Test AUC	Predicted Churners	High-Priority Contacts
SVM (RBF Kernel)	0.5997	200 / 1,000 (20%)	452 customers

1. Business Context

No-Churn Telecom is an established Indian telecom operator facing increasing competitive pressure. Despite tariff reductions and promotional offers, the company's churn rate remained above 10%. The objective of this project was to leverage Machine Learning to identify churn patterns and support customer retention strategies.

Project Goals

- Identify key variables influencing customer churn
- Predict customers likely to churn
- Generate a CHURN_FLAG variable (YES = high risk, NO = low risk)
- Provide the marketing team with a targeted retention offer list
- Provide the support team with a prioritised customer queue

2. Data Overview

Data was extracted from a MySQL database (Host: 18.136.157.135, Schema: project_telecom). The working dataset comprised 1,000 unique customers with 14 features.

Attribute	Detail
Total records	1,000 customers
Features	14 (demographic, usage, location, registration date)

Churn rate	17.9% (179 churners / 821 non-churners)
Class imbalance	82.1% No Churn vs 17.9% Churn
Missing values	None detected
Negative data_used	Present — clipped to 0 during preprocessing

3. Exploratory Data Analysis

Statistical analysis (Mann-Whitney U tests and Chi-Square tests) confirmed that feature distributions between churn and non-churn groups are closely overlapping, indicating limited separability in the dataset. Key observations included:

- Age group 45-55 showed the highest predicted churn rate (24%), while under-25s showed the lowest (15%)
- Mid-Low salary band customers churned at 36% — the most price-sensitive segment
- Reliance Jio customers showed the highest churn rate (22%) among all telecom partners
- Customer tenure, total activity score, and data usage were the most informative engineered features

4. Feature Engineering

Eight new features were derived from the raw data to improve model signal:

Feature	Description
tenure_days	Days since customer registration
total_activity	Sum of calls, SMS and data usage
call_sms_ratio	Calls made relative to SMS sent
data_per_call	Data used per call made
salary_per_dep	Estimated salary per dependent
age_group	Binned age categories (<25, 25-35, 35-45, 45-55, 55+)
salary_band	Quartile-based salary tiers (Low to High)
is_high_data_user	Binary flag: above-median data usage

5. Model Building & Results

Seven classification models were trained using a Scikit-learn Pipeline with StandardScaler and OneHotEncoder preprocessing. Class imbalance was handled using `class_weight='balanced'`. Models were evaluated using 5-fold Stratified Cross-Validation.

Model	CV AUC	Test AUC	Test Acc	Selected?
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Logistic Regression	0.5359	0.4517	50.7%	
Decision Tree	0.5080	0.4898	51.3%	
Random Forest	0.5107	0.4432	69.0%	
Gradient Boosting	0.5310	0.4906	80.3%	
K-Nearest Neighbors	0.5411	0.4036	82.0%	
Naive Bayes	0.5215	0.4574	77.7%	
SVM (RBF Kernel)	0.5156	0.5997	52.7%	BEST

The SVM model was selected as the best model based on Test AUC (0.5997) — the most reliable metric for imbalanced classification as it measures the model's ability to rank churners above non-churners regardless of class distribution.

6. CHURN_FLAG Generation

The SVM model was applied to all 1,000 customers to generate churn probabilities. Due to the conservative nature of the regularised SVM (C=0.1), probabilities were tightly clustered between 0.1472 and 0.2053. The 80th percentile (0.1864) was used as the classification threshold, aligning the predicted churn rate (20%) closely with the actual churn rate (17.9%).

Risk Tier	Customers	Prob Range	CHURN_FLAG
Critical	81	> 0.192	YES (1)
High	371	0.178 – 0.192	YES (1)
Medium	298	0.165 – 0.178	NO (0)
Low	250	< 0.165	NO (0)

7. Business Insights

By Telecom Partner

- Reliance Jio: 22% churn rate — highest among all partners
- Vodafone: 21% — second highest
- BSNL: 17% — lowest churn rate
- 7 of the top 10 highest-risk customers are on Reliance Jio or BSNL

By Age Group

- 45-55 year olds: 24% predicted churn — highest risk age segment
- 55+ customers: 22% churn rate — legacy plan holders
- Under-25s: 15% churn — most loyal segment

By Salary Band

- Mid-Low salary: 36% churn — most price-sensitive, highest priority for targeted offers
- Low salary: 3% churn — least likely to switch due to limited alternatives

Geographic Hotspot

- Himachal Pradesh appears 3 times in top 10 critical customers — a specific regional focus area

8. Business Recommendations

1. Target 452 Critical + High risk customers immediately with personalised retention offers
2. Focus retention campaigns on the Mid-Low salary band and the 45-55 age group — the two highest-risk segments
3. Investigate Reliance Jio customer experience — review tariffs and service quality to reduce the 22% churn rate
4. Improve onboarding for new customers — tenure is a key driver, meaning newer customers are more likely to churn
5. Collect richer behavioural data (e.g. complaint history, plan changes, payment delays) to improve future model AUC beyond 0.60
6. Re-train the model quarterly as new customer data is collected

9. Conclusion

This project successfully delivered all five objectives outlined in the brief. An SVM classifier with RBF kernel was identified as the best-performing model (Test AUC: 0.5997). The CHURN_FLAG variable was generated for all 1,000 customers, with 200 flagged as high risk and 452 placed on the marketing and support priority lists.

The dataset's limited feature separability — confirmed by near-identical feature means across churn groups — constrained model performance. However, the model correctly ranks customers by relative churn risk, which is the core requirement for actionable business outputs. With richer behavioural data and continued model iteration, predictive accuracy can be meaningfully improved.

Deliverables: Jupyter Notebook (.ipynb) | Predictions CSV (churn_predictions.csv)
